# An Update to the Gravitational Wave Galaxy Catalogue for the Rapid Search of Electromagnetic Counterparts from Gravitational Waves

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## Abstract

The Laser Interferometer Gravitational-Wave Observatory (LIGO), is a large scale Michelson interferometer which aim is to directly detect gravitational waves predicted by Albert Einstein in 1916. The gravitational waves sources that LIGO expects to detect consist of the merging of two neutron stars or a neutron and black hole at cosmic distances. It is anticipated that the coalescence of these compact objects may also produce a short lived gamma ray burst (GRB). Unfortunately, LIGO is not accurate in localizing the points in the sky where the gravitational sources are be found. The best that LIGO can do is to constrain a gravitational wave source in an area of approximately 100 square degrees where literally millions of electromagnetic sources can be observed. So its seems that the simultaneous detection of gravitational and electromagnetic waves coming from a source is highly unlikely. To facilitate the task of searching for electromagnetic counterparts coming from gravitational waves a list of galaxies within 100 Mpc has been compiled, the gravitational waves galaxy catalog (GWGC). LIGO is now being design to improve its sensitivity by a factor of 10 which means that it would be able to observe neutron star/ neutron star coalescence to distances of 300 Mpc and neutron star/ black hole coalescence up to 650 Mpc. Therefore, GWGC must be updated to contain galaxies at these distances. We present the results of the updates made to GWGC.

## Introduction

In 1916, Albert Einstein predicted that the acceleration of heavy masses will produce ripples in space-time known as gravitational waves (GWs). Such ripples will contract and expand the distance between two fixed points in space-time. Based on such principle a network of GWs detectors (LIGO, VIRGO, and GEO) has been built. In the ideal scenario, the GWs sources observed at these detectors will be confine in a region of the sky and quickly a telescope will be assign to observe it. What we expect to observe is a short lived electromagnetic (EM) signal, extragalactic in origin.

## The Gravitational Wave Galaxy Catalog (GWGC)

A major problem that arises with the previously mention scenario is that the area of the sky these detectors confine the GW s sources is extremely large ($\sim 100$ deg$^2$), where simultaneously many transients can be observed. A solution to this problem is to constrain EM transients follow ups by the horizon distance LIGO & VIRGO places for most of its sources ($\sim 40$ Mpc). For this reason, the Gravitational Wave Galaxy Catalogue (GWGC) which is a list of galaxies that are within 100 Mpc [1], has been compiled . Figure 1, 2, and 3 describe the completeness of the catalog, the advantage of having a catalog, and the blue light luminosity distribution respectively.

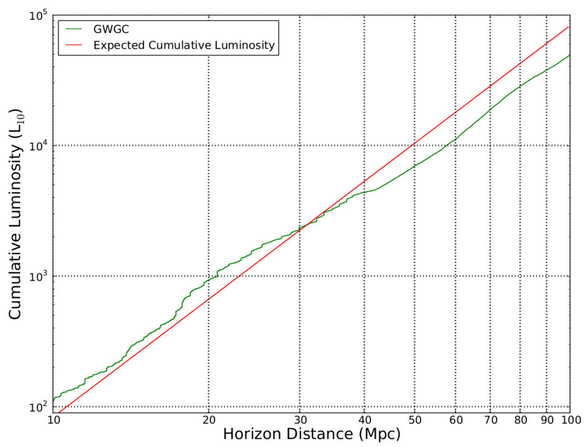



Figure 1: To get a sense of how complete GWGC is, we compare GWGC cumulative luminosity (shown in green) with the expected cumulative luminosity (shown in red). As you can see GWGC begins to loose completeness after 30 Mpc where GWGC cumulative luminosity begins to fall below the expected cumulative luminosity.

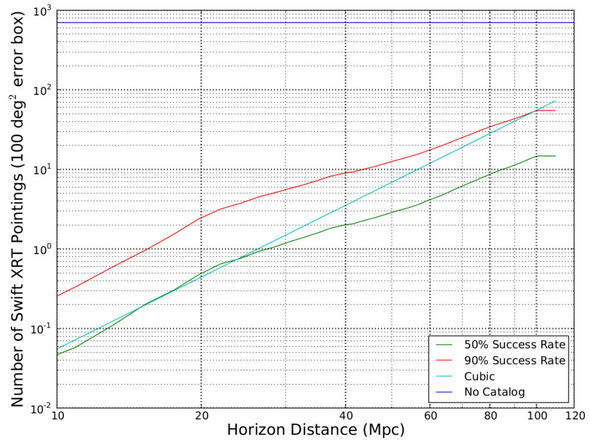



Figure 2: To see the convenience of having GWGC, we compare the number of Swift XRT pointings needed to cover 100 deg$^2$ of sky VS the number of Swift XRT pointings needed to observe 50% & 90 % of the blue light luminosity within 100 deg$^2$ of sky has a function of distance. Blue light luminosity is an indicator of the star formation occuring in a galaxy. From empirical data it seems that short lived gamma ray burst (GRB), the expected EM radiation originating from a GW source, are more like to occur at galaxies with higher blue light luminosity.

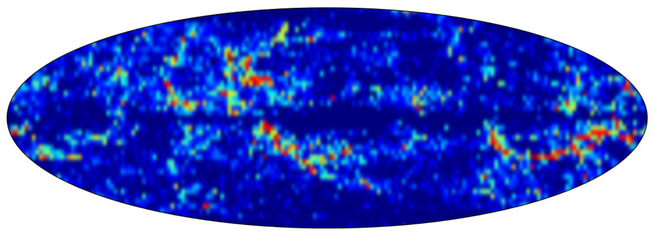


Figure 3: The blue light luminosity distribution of the galaxies in GWGC. Regions colored in blue represent lower accumulation of blue light luminosity and regions colored in red amount to high concentrations of blue light luminosity.

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## Cross Correlation of GWGC with the Wide-field Infrared Survey Explorer (WISE)

Recent observations show evidence that short GRBs generally arise in intermediate or old stellar populations. For this reason, the K-band magnitude, an indicator of the mass of the galaxy, needed to be added to GWGC galaxies. To do so, we cross correlated galaxies in GWGC with galaxies in the WISE catalog. Of the 56,970 galaxies in GWGC, 34,597 had a unique match with one of the 22,727,272 galaxies in the WISE catalog.

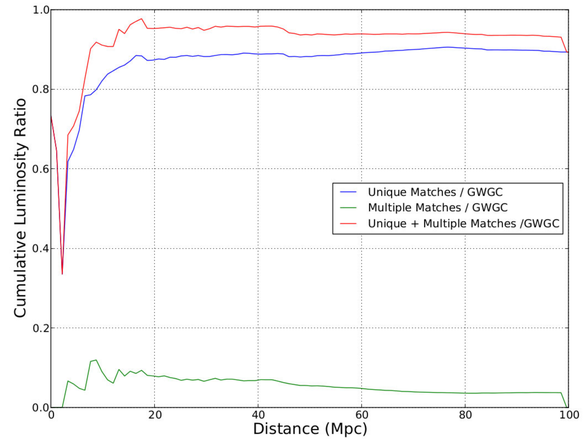



Figure 4: At first sight it seems that the cross correlation between GWGC and WISE performed poorly. To investigate, we compute the ratio between the cumulative blue luminosity of galaxies in GWGC with a unique match in WISE and the cumulative blue luminosity of all GWGC galaxies as a function of distance (shown in blue). The closer the ratio is to one the more blue light luminosity we recovered.

## Conclusion

Since about 40 % of GWGC galaxies don't have a K-band magnitude, we need to cross correlate with other catalogs in order to attain them. Furthermore, GWGC must be expanded to included galaxies at greater distances because by 2015 the GW sources LIGO expects to detect will be observable to distances of 650 Mpc.

## References

[1] D. J. White, E. J. Daw, and V. S. Dhillon, Classical and Quantum Gravity **28**, 085016 (2011).